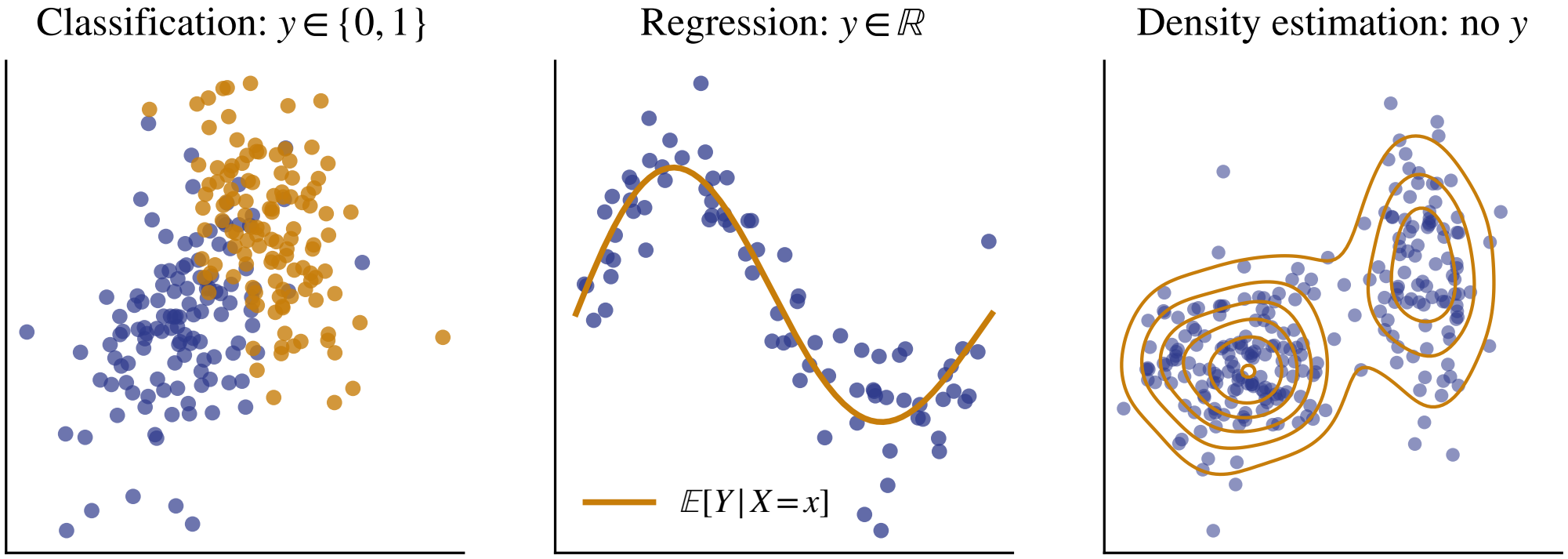
Classification: $y \in \{0, 1\}$
Regression: $y \in \mathbb{R}$
Density estimation: no y
$\mathbb{E}[Y \mid X = x]$